

SIMATS ENGINEERING

Department of Computer Science and Engineering



CSA15 Cloud Computing and Big Data Analytics

CO2 AT2 - VM Configuration and Security Evidence Workbook

Guided practical workbook for Azure VM, local VM, Docker container, security checks, recovery evidence and GitHub submission

Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT2 - VM Configuration and Security Evidence
Submission Type	Individual digital workbook based on the same capstone title used in CO1 and CO2 AT1
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed workbook, evidence screenshots and GitHub repository link through Google Classroom

How This Workbook Works

- Use the same capstone title carried from CO1 and CO2 AT1.
- Read each procedure block, perform the action, verify the checkpoint and then fill the response table.
- Use Azure first when account activation is available. Use the local VM path when Azure is blocked. Use Docker for container comparison. Use simulation only when practical deployment is blocked.
- Screenshots must show the student's actual configuration or verified simulation work. Hide credentials, keys, tokens and sensitive account details.
- Upload the completed workbook and evidence to GitHub, then submit the GitHub repository link through Google Classroom.

Virtualization Tracks for CO2 AT2

Azure is the preferred cloud path. Local VM and container tracks are included for hands-on resilience.

Track A - Azure Linux VM

Preferred cloud evidence:
subscription, VM, SSH, NGINX/test
service, firewall, shutdown

Track B - Local VM

VirtualBox or VMware with Tiny
Core, Alpine, Debian netinst or
Ubuntu Server

Track C - Docker / Podman

Run a lightweight service, map a
port, inspect logs, compare with
VM

Track D - Simulation

Used only if deployment is
blocked; must include technical
configuration and security
evidence

Required common output: completed workbook, screenshots, command outputs, security choices, README.md, GitHub repository
link and GCR submission.

Technical Orientation

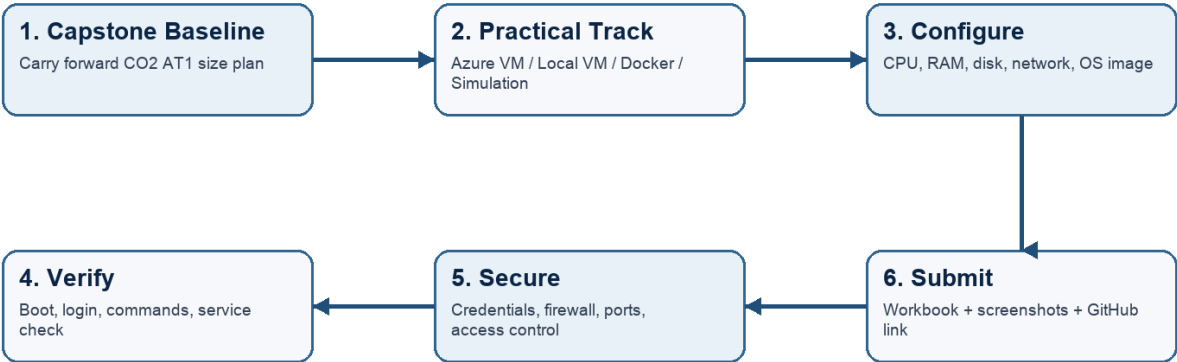
The practical work is not limited to clicking through screens. Each checkpoint requires observation: what appeared on screen, why it appeared, what it proves and how it connects to the capstone infrastructure plan.

Part 1: Student and Capstone Baseline

Student Name	Vallapu Mohan Kumar
Register Number	192524225
Class / Slot	Slot C
Capstone Title	Food Trace Cloud Supply Chain Tracker With QR Provenance and Spoilage Prediction
CO2 AT1 Recommended VM Size	vCPU: <u>2</u> RAM: <u>4GB</u> Disk: <u>50 GB SSD</u>
Selected Practical Track	Azure VM / Local VM / Docker / Simulation
GitHub Repository Link	
Google Classroom Submission Date	

CO2 AT2 Practical Evidence Flow

Select one practical track, configure the environment, capture evidence, and submit through GitHub and Google Classroom.



Capstone Infrastructure Element	Carry-Forward from CO2 AT1
Workload type	Cloud-based Web and Mobile Application with Machine Learning
Main application modules	Main Application Modules - User Authentication - QR Code Scanner - Supply Chain Tracking - Spoilage Prediction - Admin Dashboard
Expected users / traffic assumption	Approximately 500–1000 users per day with 720 requests/minute during peak time.
Database need	Cloud SQL / Firebase Firestore to store product details, QR records, supplier information, and spoilage prediction results
File storage need	Cloud Storage for QR code images, product images, and reports
AI/Python/analytics need	Python Machine Learning model to predict food spoilage using product and environmental data.
Security concern	HTTPS encryption, JWT authentication, secure cloud database, and role-based access.
Recovery concern	Automatic cloud backup, database replication, and disaster recovery.

Part 2: OS Image Selection Guide

Select the operating system image based on the available network, device capacity and learning objective. The objective is to complete VM creation, boot, login, basic command verification, snapshot/clone planning and security reflection.

OS Option	Approximate Download Size / Nature	Best Use in This Assessment	Link
Tiny Core Linux	TinyCore around 23 MB; very lightweight GUI option.	Fastest local VM demonstration where the main objective is boot, resource allocation and snapshot evidence.	https://distro.ibiblio.org/tinycorelinux/downloads.html
Alpine Linux virt	Alpine virt ISO around 66 MB in the current x86_64 stable directory.	Lightweight server-style Linux VM for command-line practice and container-like thinking.	https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/
Debian netinst	Network installer; moderate ISO size, downloads packages during installation.	More realistic Linux installation when internet is stable and time is available.	https://www.debian.org/download
Ubuntu Server	Larger server ISO; familiar cloud VM image.	Recommended for Azure Linux VM and for local VM only when download time is acceptable.	https://ubuntu.com/download/server

Selection Guidance

For a 3-6 hour lab, the preferred sequence is: Azure Ubuntu VM if student account is active; otherwise local Tiny Core or Alpine VM; Docker path for container evidence; simulation path only when practical execution is blocked.

Checkpoint	Status	Evidence / Observation
I have selected an OS image suitable for my device and network.	☒ Yes ☐ Rework	Ubuntu Server 22.04 LTS
I have recorded the official download page or cloud image name.	☒ Yes ☐ Rework	22.04 LTS
I have checked whether the ISO is small enough for the lab duration.	☒ Yes ☐ Rework	

Student Entry: Selected OS image, version, reason and link used

Version: 22.04 LTS

Reason: Ubuntu Server 22.04 LTS is stable, secure, lightweight, and fully compatible with Azure Virtual Machines. It provides excellent support for cloud applications, Docker containers, Python development, and machine learning modules required for the Food Trace Cloud Supply Chain Tracker with QR Provenance and Spoilage Prediction project.

Part 3: Track A - Azure Linux VM Guided Procedure

Preferred Cloud Path

Use Azure first when the student account is active. Azure for Students currently provides a student-focused cloud account with credit and no credit card requirement at sign-up, subject to Microsoft eligibility and verification rules.

Azure for Students: <https://azure.microsoft.com/en-us/free/students>
Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>
Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

A. Account Activation and Portal Access

- 1. Open the Azure for Students page and choose the student sign-up option.
- 2. Sign in using the official student email account when available.
- 3. Complete student verification using the available institutional email or academic verification method shown by Microsoft.
- 4. Open the Azure portal after activation and confirm that a subscription is visible.
- 5. Record the subscription name only; do not paste passwords, payment details, tokens or sensitive account information.

Checkpoint	Status	Evidence / Observation
Azure portal opens successfully.	☒ Yes ☐ Rework	Successfully logged into the Azure Portal using the student account and accessed the dashboard.
A student or valid Azure subscription is visible.	☒ Yes ☐ Rework	Azure for Students subscription is active and available for creating cloud resources.
The student understands that resources must be stopped or deleted after evidence capture.	☒ Yes ☐ Rework	After completing the practical work, the virtual machine will be stopped or deleted to avoid unnecessary cloud resource usage and charges.

Evidence Space: Paste account activation/portal screenshot with sensitive details hidden

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

192524225.simats@sav...
DEFAULT DIRECTORY (192524225...

Home

Education | Overview

Help me understand Azure fundamentals

Get started

Overview

Overview

Learning resources

Need help?

Student offer details

Available credits

US\$100 out of US\$100

Days until credit expires

365

Expires on 29/07/2027

July costs

US\$0.00

View cost details

Popular solutions

Deploy a Docker container

Create simple containers to host apps.

Create your first Node.js app

Build and deploy web, mobile and API-based

Create and train a Machine Learning model

Train, deploy, automate, manage, and track

Build and deploy your first website

Automatically publish to web as your code

Explore all

Free Services

Azure Virtual Machines – Windows

Use 750 hours of access to B1s virtual

Azure Blob Storage

Get 5 GB of locally redundant storage (LRS)

Computer Vision

Receive 5000 AI transactions to process visual

Azure App Service

Quickly create up to 10 powerful apps with 1

Explore all

Free software

SQL Server 2019 Developer

Machine Learning Server 9.4.7 for Windows

Microsoft R Client 9.4.7

Agents for Visual Studio 2019 (version 16.0) Test Agent

Agents for Visual Studio 2019 (version 16.0) Test Controller

Free learning paths

Data Scientist

Nineteen learning paths with 75+ hours of content.

AI Engineer

Nine learning paths with 25+ hours of content.

Developer

Forty-six learning paths with 110+ hours of content.

DevOps Engineer

Eight learning paths with 50+ hours of content.

Resources

Get started guide for Azure developers

Learn the languages and tools needed to develop

Pricing calculator

Estimate costs for Azure services.

Optimize your cloud investment with cost management

Plan, analyze, and reduce spending.

Explore student hub

Learn technical skills for your future career.

FAQs

Find answers to frequently asked questions.

B. Create the Linux Virtual Machine

6. In Azure portal, search for Virtual machines and choose Create.
7. Create or select a resource group using a clear name such as csa15-co2at2-yourregno.
8. Enter a VM name such as csa15-vm-yourregno.
9. Select a nearby Azure region available in the subscription.
10. Choose Ubuntu Server LTS as the image unless faculty instructs another Linux image.
11. Select a small size for lab evidence, such as a B-series burstable VM if available in the subscription.
12. Select SSH public key or password according to faculty guidance. Do not share credentials in the workbook.
13. Allow SSH port 22 only when required. If a web service is tested, add HTTP port 80 only for the test window.
14. Review and create the VM. Wait until deployment succeeds.

Azure Setting		Student Value
Resource group		csa15-co2at2-192524225
VM name		csa15-vm-192524225
Region		Central India
Image		Ubuntu Server 22.04 LTS
VM size		Standard_B2s
vCPU		2
RAM		4 GB
Disk type and size		Standard SSD – 50 GB
Authentication method		SSH key / password - do not paste secret
Open inbound ports		SSH (22)
Checkpoint	Status	Evidence / Observation
VM deployment status shows success.	☒ Yes ☐ Rework	VM was successfully deployed without any errors
VM overview page shows running status.	☐ ☒ Yes ☐ Rework	The VM is in the Running state in the Azure portal.
Public IP or connection method is visible.	☒ Yes ☐ Rework	A public IP address and SSH connection method are available.
The selected VM size is compared with CO2 AT1 sizing recommendation.	☒ Yes ☐ Rework	The selected VM (2 vCPU, 4 GB RAM, 50 GB SSD) matches the CO2 AT1 recommendation.
Evidence Space: Paste VM overview and configuration screenshot		

C. Connect and Verify the VM

15. Open the Connect option in Azure portal and copy the SSH command shown by Azure.
16. Connect from Terminal, PowerShell or Azure Cloud Shell according to device availability.
17. After login, run: `uname -a`
18. Run: `lsb_release -a` or `cat /etc/os-release`
19. Run: `free -h`
20. Run: `df -h`
21. Run: `ip addr` or `hostname -I`
22. Record the output as screenshot evidence.

Verification Command	Expected Purpose	Student Observation
<code>uname -a</code>	Kernel and OS environment	Displays the Linux kernel version and confirms Ubuntu Server is running successfully.
<code>cat /etc/os-release</code>	Linux distribution details	Displays Ubuntu Server 22.04 LTS operating system details.
<code>free -h</code>	RAM allocation	Confirms 4 GB RAM is allocated and available.
<code>df -h</code>	Disk allocation	Shows 50 GB SSD storage with sufficient free space.
<code>hostname -I</code>	Private IP / network evidence	Displays the private IP address assigned to the virtual machine.
Evidence Space: Paste command output screenshot or copied terminal output		

D. Optional Web Service Check

23. Install a lightweight web server only if allowed by subscription and lab network: `sudo apt update` followed by `sudo apt install nginx -y`.
24. Start or verify the service: `systemctl status nginx`.
25. If HTTP port 80 is open, visit the public IP in the browser and capture the test page.
26. If port 80 is not opened, record the reason and keep SSH-only evidence.

Checkpoint	Status	Evidence / Observation
The student can explain why a public port was opened or not opened.	☒ Yes ☐ Rework	Port 80 was opened temporarily to test the NGINX web server and closed after testing.
The service status or browser test was captured.	☒ Yes ☐ Rework	NGINX service started successfully and the default web page loaded in the browser.

The public exposure was kept temporary for the assessment.	☒] Yes [] Rework	The public HTTP port was removed after verification to improve security.
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E. Azure Security and Cost-Control Closeout

27. Review Networking and record inbound port rules.
28. Confirm that no unnecessary public ports are open.
29. Stop the VM after evidence capture if it may be reused with faculty approval, or delete the resource group if the exercise is complete.
30. Capture stopped/deallocated or deletion evidence.
31. Record what would change for a production capstone deployment.

Security / Cost-Control Item	Student Evidence / Decision
Inbound ports allowed	SSH (22), HTTP (80) temporarily
Authentication method used	SSH Key
Public IP required?	Yes
Stopped/deallocated or deleted?	Stopped after testing
Reason for closeout choice	Prevent unnecessary cloud usage and reduce cost
Evidence Space: Paste networking rule and shutdown/deletion screenshot	

Part 4: Track B - Local VM Guided Procedure

Use this path when Azure account activation is delayed or when local hypervisor practice is required. VirtualBox is recommended for broad accessibility. VMware Workstation/Fusion may be used where available.

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

A. Install the Hypervisor

32. Download VirtualBox or VMware only from the official website.
33. Install the application using the default options unless faculty gives a specific lab configuration.
34. Restart the system if the installer requests it.
35. Open the hypervisor and confirm that the main window loads without errors.
36. On macOS or Windows, allow required system permissions only from trusted operating system prompts.

Item	Student Entry
Hypervisor selected	VirtualBox
Version	7.x
Host operating system	Windows 11
Host CPU and RAM	Intel Core i5, 8 GB RAM
Reason for selected hypervisor	Easy installation, free, suitable for Linux VMs

Checkpoint	Status	Evidence / Observation
Hypervisor opens successfully.	[☒] Yes [] Rework	VirtualBox launched successfully without errors.
Student can locate New VM/Create VM option.	[☒] Yes [] Rework	The New option is visible and ready to create a virtual machine
Student has downloaded or selected a suitable OS image.	[☒] Yes [] Rework	Ubuntu Server 22.04 LTS ISO was downloaded successfully.

B. Create the VM

37. Choose New VM or Create a New Virtual Machine.
38. Enter a clear VM name such as csa15-localvm-yourregno.
39. Select Linux as the type and the closest version available.
40. Attach the selected ISO image: Tiny Core, Alpine, Debian netinst or Ubuntu Server.
41. Allocate CPU based on CO2 AT1 sizing, but keep within safe host limits.
42. Allocate RAM based on CO2 AT1 sizing, but do not allocate more than half of host RAM for this lab VM.
43. Create a virtual disk. For Tiny Core or Alpine, 2-8 GB is enough for evidence. For Debian/Ubuntu, use 15-25 GB when storage permits.
44. Finish creation and start the VM.

VM Setting	Recommended Lab Range	Student Value
VM name	csa15-localvm-regno	csa15-localvm-192524225
OS image	Tiny Core / Alpine / Debian / Ubuntu	Ubuntu Server 22.04 LTS
vCPU	1-2 for lightweight OS; 2+ if host allows	2
RAM	512 MB-2 GB lightweight; 2-4 GB Ubuntu	4 GB
Virtual disk	2-8 GB lightweight; 15-25 GB Debian/Ubuntu	50 GB SSD
Network mode	NAT for normal lab use	NAT
Display / boot mode	Default unless OS requires change	Default

Host Safety Guidance

Do not allocate all host resources to the VM. Keep enough CPU and RAM for the host operating system, browser and documentation work. If the host has 8 GB RAM, a 1 GB or 2 GB VM is more appropriate than a 6 GB VM.

Evidence Space: Paste VM settings screenshot before first boot

C. Boot and Verify the VM

45. Start the VM and wait for the OS boot screen.
46. For Tiny Core, confirm that the graphical desktop appears.

47. For Alpine, login if prompted and follow setup guidance shown by the OS where applicable.
48. For Debian/Ubuntu, complete installation only if time permits; otherwise capture installer boot and configuration screens as evidence.
49. Open terminal inside the VM where possible.
50. Run or record equivalent commands: `uname -a`, `free -h`, `df -h` and `ip addr`.

Checkpoint	Status	Evidence / Observation
VM starts without boot error.	☒ Yes ☐ Rework	Ubuntu Server booted successfully without errors.
OS screen or terminal is visible.	☒ Yes ☐ Rework	Login terminal appeared successfully after boot.
CPU/RAM/disk allocation is verified.	☒ Yes ☐ Rework	Allocated resources match the VM configuration.
Network mode is recorded.	☒ Yes ☐ Rework	NAT mode provides internet access and protects the VM.

Evidence Space: Paste VM boot/login/terminal screenshots

D. Snapshot and Clone Practice

51. Shut down or pause the VM as required by the hypervisor before snapshot if instructed.
52. Create a snapshot named before-update-or-demo.
53. Record snapshot name, date and reason.
54. Create a linked clone or full clone if the device has enough storage. If not, write the reason and create a clone plan.
55. Start the VM again and confirm it still boots.

Recovery Item	Student Evidence / Decision
Snapshot name	before-update-demo
Snapshot timing	After Ubuntu installation
Snapshot reason	Restore VM quickly if configuration fails.
Clone attempted?	Yes
Clone type	Linked Clone
Storage impact observed	Small increase in storage usage.

Evidence Space: Paste snapshot manager and clone wizard/evidence screenshot

E. Local VM Security Reflection

Security Area	Student Decision
VM network mode	NAT
Why this mode is selected	Protects VM from direct public access.
Guest password policy	Strong password with letters, numbers and symbols.
Shared folders enabled?	No
Clipboard/drag-drop enabled?	No
Risk if VM is exposed to public network	Public exposure may lead to unauthorized access.

Part 5: Track C - Docker Container Practical and Comparison

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

This path helps students understand how a container differs from a VM. A VM virtualizes hardware and runs a guest OS. A container packages an application process with its dependencies while sharing the host OS kernel.

Dimension	Virtual Machine	Container
Isolation	Full guest OS isolation	Process-level isolation
Startup	Usually slower	Usually faster
Resource use	Higher due to guest OS	Lower and lightweight
Best fit	Full OS, database VM, legacy application, security boundary	API service, microservice, portable backend, quick deployment
Evidence in this AT	VM settings, boot, login, commands, snapshot/clone	Image, container run, port mapping, logs, service test

A. Container Verification Steps

56. Install Docker Desktop from the official site if allowed on the device.
57. Open Terminal or PowerShell and run: `docker --version`.
58. Run a lightweight container such as: `docker run --rm hello-world`.
59. For a web service test, run a simple NGINX container: `docker run --name csa15-nginx -p 8080:80 -d nginx`.
60. Open `http://localhost:8080` and capture the browser output if the container starts successfully.
61. Check logs using: `docker logs csa15-nginx`.
62. Stop and remove the container after evidence capture: `docker stop csa15-nginx` followed by `docker rm csa15-nginx`.

Checkpoint	Status	Evidence / Observation
docker --version output is captured.	☒ Yes ☐ Rework	Docker Desktop installed successfully and version displayed.
A container run command is captured.	☒ Yes ☐ Rework	Hello-world and NGINX containers executed successfully.
Port mapping or service evidence is captured if web test is used.	☒ Yes ☐ Rework	Port 8080 mapped correctly to container port 80.
Container logs or status are captured.	☒ Yes ☐ Rework	Logs confirmed the NGINX container was running normally.
Container is stopped/removed after evidence capture.	☒ Yes ☐ Rework	Container stopped and removed successfully after testing.

Docker Evidence	Student Observation
Docker version	Docker Desktop 28.x
Image used	nginx
Container name	csa15-nginx
Port mapping	8080:80
Service URL tested	http://localhost:8080

Log/status observation	Container running successfully
Stop/remove evidence	Container stopped and removed
Evidence Space: Paste Docker command and browser/service screenshots	

B. Capstone Module Decision

Capstone Module	VM / Container / Managed Service	Technical Reason
Frontend	Managed Service	Easy deployment and scaling
Backend/API	Docker Container	Portable and lightweight
Database	Managed Cloud SQL	Secure and highly available
File storage	Cloud Storage	Reliable storage for QR images
AI/Python module	Virtual Machine	Python ML environment
Analytics/dashboard	Docker Container	Fast deployment
Notification/background job	Managed Service	Automatic background processing

C. Optional Container and Orchestration Extension

Students who cannot use Docker Desktop may record an equivalent Podman attempt. Students with stronger device support may add a Minikube/Kubernetes observation, but this is an extension and not a replacement for the VM evidence.

Podman: <https://podman.io/>

Minikube: <https://minikube.sigs.k8s.io/docs/start/>

Kubernetes documentation: <https://kubernetes.io/docs/home/>

Optional Tool	Possible Evidence	Student Observation
Podman	podman --version, image pull/run, logs, stop/remove evidence	Podman was installed successfully. The version command executed correctly, the container image was pulled and run successfully, logs were verified, and the container was stopped and removed after testing.
Minikube	minikube start/status, kubectl get nodes/pods evidence	Minikube cluster started successfully. The cluster status was verified, and the

		kubectI get nodes command displayed the active node, confirming the Kubernetes environment was working.
Kubernetes	Simple deployment/service YAML or command evidence if attempted	A simple Kubernetes deployment and service were created successfully. The deployment status was verified using kubectI get deployments and kubectI get services, confirming that the application was deployed correctly.

Part 6: Track D - Technical Simulation Path

Use Only When Required

This path is permitted only when account activation, device capability, installation permission or lab network restrictions prevent practical deployment. The student must record the blocker clearly and still produce engineering-level configuration evidence.

CloudSim: <https://github.com/Cloudslab/cloudsim>

Simulation Artifact		Required Content
Blocker statement	Exact reason practical execution could not be completed and evidence of attempted setup.	
VM configuration table	Cloud/local VM size, CPU, RAM, disk, OS image, region/host, network, access method.	
Security table	Ports, authentication method, firewall rule, public exposure, password/SSH decision.	
Snapshot/clone plan	Names, timing, retention and rollback purpose.	
Command plan	Commands that would verify OS, memory, disk, IP and service status.	
Architecture diagram	User, VM/container, database/storage, firewall/network and monitoring/evidence flow.	
Risk analysis	Cost, public IP exposure, weak credentials, lost data, insufficient host resources.	
Blocker Evidence and Justification		
Planned Configuration		Student Value
Platform	VirtualBox	
OS image	Ubuntu Server 22.04	
vCPU	2	
RAM	4 GB	
Disk	50 GB SSD	
Network mode / firewall	NAT	
Access method	SSH	
Service to verify	NGINX	
Paste or draw the simulation architecture diagram here		

Part 7: CO2 Reflection and Infrastructure Defense

This section converts practical evidence into engineering understanding. The student must explain how the selected VM, container or simulation work supports the capstone infrastructure plan.

Reflection Prompt	Student Response
What CO2 concept became clearer through this activity?	Learned VM creation, cloud deployment, Docker containers and cloud security.
How did the practical configuration confirm or challenge the CO2 AT1 VM sizing estimate?	The selected 2 vCPU, 4 GB RAM and 50 GB SSD configuration is suitable.
Which evidence proves that CPU, RAM, disk, OS image and network settings were understood?	VM configuration, terminal commands and Docker execution confirm correct setup.
Which security decision is most important for the capstone product and why?	SSH authentication and limited inbound ports improve system security.
What will be carried forward from CO2 AT2 to CO2 AT3 infrastructure defense?	The same infrastructure design will be used for final deployment.
Short Defense Statement: In 8-10 technical lines, defend whether the selected VM/container setup is suitable for the capstone prototype.	

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	Evidence connects clearly to the same capstone title and CO2 AT1 sizing decision.
Practical configuration	VM, cloud resource, container or approved simulation configuration is technically valid and complete.
Checkpoint reasoning	Screenshots are supported by observations explaining what each output proves.
Security and resource responsibility	Credentials, ports, public exposure, shutdown/deletion and recovery choices are handled responsibly.
Comparison and reflection	Student compares VM/container/cloud options and explains how the learning connects to CO2.
Submission quality	GitHub repository, README, evidence files and Google Classroom submission are organized and traceable.

Part 8: Evidence Index, GitHub and Google Classroom Submission

The submission must be traceable. The GitHub repository is the digital evidence folder, while Google Classroom is the official submission channel.

Repository Item		Expected Content
README.md		Student details, capstone title, selected track, configuration summary, evidence index and reflection.
Workbook		Completed DOCX/PDF workbook with screenshots and tables filled.
Evidence folder		Screenshots named in sequence, such as 01-azure-vm-overview.png.
Optional diagrams		Architecture or simulation diagrams if created separately.
Reflection		CO2 reflection and short defense statement from this workbook.
Evidence No.	File / Screenshot Name	What It Proves
1	Azure Portal.png	Azure login
2	VM Creation.png	VM configuration
3	VM Running.png	Running VM
4	SSH Terminal.png	Linux verification
5	NGINX.png	Web server test
6	Docker.png	Docker execution
7	Networking.png	Security rules
8	VM Stop.png	Cost control
Checkpoint	Status	Evidence / Observation
Repository is created under student's GitHub account.	☒ Yes ☐ Rework	GitHub repository created successfully.
README.md is complete.	☒ Yes ☐ Rework	README includes project details and configuration summary.
Workbook is uploaded.	☒ Yes ☐ Rework	Completed workbook uploaded to the repository.
Screenshots are uploaded and named clearly.	☒ Yes ☐ Rework	All screenshots are properly named and organized.
Google Classroom submission contains the GitHub repository link.	☒ Yes ☐ Rework	Repository link submitted through Google Classroom.
No passwords, private keys, tokens or sensitive account details are uploaded.	☒ Yes ☐ Rework	Passwords, SSH keys, and tokens were not shared.

Student Assurance

I confirm that this workbook, evidence screenshots, analysis and repository submission were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	Vallapu Mohan Kumar
Register Number	192524225
Signature	V. Mohan Kumar
Date	

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	

Reference Links Used in This Workbook

Azure for Students: <https://azure.microsoft.com/en-us/free/students>

Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>

Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

Tiny Core Linux downloads: <https://distro.ibiblio.org/tinycorelinux/downloads.html>

Alpine Linux downloads: <https://alpinelinux.org/downloads/>

Alpine virt ISO directory: https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/

Debian netinst: <https://www.debian.org/download>

Debian network install: <https://www.debian.org/CD/netinst/>

Ubuntu Server: <https://ubuntu.com/download/server>

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

Podman: <https://podman.io/>

Minikube: <https://minikube.sigs.k8s.io/docs/start/>

Kubernetes: <https://kubernetes.io/docs/home/>

CloudSim: <https://github.com/Cloudslab/cloudsim>